

PICOFLOW · REQUIREMENTS

PicoFlow requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/picoflow		
Source	docs/03-requirements.md @ a4c970d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

PicoFlow requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users or at a site, and will be checked by calculation at TRL 3. The status column compares each target with the first-order estimates in PCF-PRC-001; every status is an estimate.

The **design point** used throughout is 2.0 m of gross head (forebay water surface to nozzle centerline) and 10 L/s through two nozzles, with a 20 m run of 110 mm PVC penstock.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (estimate)
R1	Operating head range	Generates from 1.0 to 3.0 m gross head with the same runner, changing only nozzle inserts	Performance calculation across the range	Met by design
R2	Flow range	5 to 15 L/s through interchangeable nozzle inserts, one or two jets	Nozzle sizing calculation	Met by design
R3	Output at the design point	80 W or more DC into the battery at 2.0 m and 10 L/s	Efficiency chain calculation; later bench test	Met, about 90 W, thin margin
R4	Output at the lowest head	30 W or more DC into the battery at 1.0 m and 7 L/s	Efficiency chain calculation	At risk , about 30 W
R5	Water-to-wire efficiency	40 % or more at the design point, from gross hydraulic power to DC into the battery	Calculation; later test	Met, about 46 %
R6	Daily energy	1.8 kWh or more per day at the design point, running 24 h	Calculation	Met, about 2.2 kWh
R7	Battery charging and load control	Charges a 12 V nominal battery; holds the runner within 10 % of its best speed while charging; diverts 100 % of output to the dump load within 1 s when the battery is full or disconnected	Controller design review; later bench test	Unverified
R8	Runaway and touch voltage	Every part survives indefinite runaway at 3.0 m head with no damage; no exposed conductor above 60 V DC in any state, including runaway	Speed and voltage calculation; design review	Not met , runaway open-circuit voltage about 83 V
R9	Printable parts	Runner and nozzles print on a 220 x 220 x 250 mm build volume in PETG or nylon; runner prints in 24 h or less	Model check; slicer estimate	Met by geometry (200 mm runner)
R10	Debris tolerance	Intake screen aperture 6 mm or less; smallest nozzle 20 mm or more; screen self-cleans by overflow	Design review	Met by design
R11	Serviceability	Runner or nozzle inserts replaced with hand tools in 30 min or less; bearings in 60 min or less; bearings and generator above the spray zone	Design review; later timed trial	Met by layout, unverified
R12	Service life	Bearing L10 life 20,000 h or more; runner 1 year (8,760 h) or more before replacement	Bearing life calculation; material review	At risk , printed runner creep and erosion unverified
R13	Setting height	Nozzle centerline 0.3 m or less above normal tailwater, so little head is lost below the jet	Model check	Not met , about 0.44 m in the massing model
R14	Portability	Heaviest single item 15 kg or less; turbine unit 25 kg or less, carried to site by two people	Mass estimate	Met, about 20 kg (estimate)
R15	Cost	Turbine kit (all items except penstock and battery) \$350 or less in parts	Priced BOM (bom/ bom.csv)	Not met , about \$424 with a new generator;

ID	Requirement	Target	Verification (TRL 3 or later)	Status (estimate)
				about \$354 with a salvaged one
R16	Stream protection	Takes no more than half of the dry-season stream flow and returns all water to the same stream; intake passes fish-safe overflow	Site survey; design review	Site-dependent

■ Requirements not met or at risk

- **R8 (touch voltage) not met.** At 3.0 m head a runner with no load runs at about 830 rpm, which drives an estimated 83 V open circuit from a generator sized for 12 V charging. Options are a voltage clamp on the dump load, an enclosed DC side rated for 100 V, or both.
- **R13 (setting height) not met.** The frame in the massing model puts the nozzles about 0.44 m above the ground, which wastes about 18 % of a 2.44 m site drop. The frame must be lowered or the housing sunk into the tailrace, while keeping the runner above flood tailwater.
- **R15 (cost) not met.** About \$424 with a new low-speed BLDC generator, about 21 % over. A salvaged direct-drive washing machine motor brings the kit to about \$354, about at budget within the accuracy of these estimates.
- **R4 (1 m output) and R12 (runner life) at risk.**

■ Assumptions

- Penstock friction loss about 10 % of gross head (110 mm PVC, 20 m, 10 L/s). Shorter or larger pipes lose less.
- Nozzle velocity coefficient 0.97.
- Runner jet-to-shaft efficiency 75 % for a first printed design, below the 87 to 91 % reached in laboratory work on an optimized low-head Turgo ([Williamson et al., 2013](#)).
- Generator efficiency 80 % at partial load and low speed; rectifier and MPPT converter 90 %.
- Generator constant about 10 rpm per volt of rectified DC (typical of low-speed direct-drive BLDC motors; to be confirmed for the chosen unit).